

ANCHORAGE HARBOR, AK, USA

WMO: 997381

Lat: 61.238N

Lon: 149.890W

Elev: 10

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.4	(c) -0.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 22.7	(k) 22.9	(l) 18.8	(m) 23.1	(n) 4.3	(o) 110	(p) 0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.7	(c) 66.6	(d) N/A	(e) 64.6	(f) N/A	(g) 62.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.9	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.4	(b) 16.0	(c) 12.8	(e) -5.9	(f) 72.7	(g) 4.4	(h) 4.1	(i) -9.1	(j) 75.6	(k) -11.6	(l) 77.9	(m) -14.1	(n) 80.2	(o) -17.2	(p) 83.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	18.3	21.2	26.2	36.2	47.6	54.9	58.3	56.9	49.8	38.5	24.2	20.3
	DBStd	16.37	11.14	9.11	8.08	5.42	4.98	3.74	3.06	2.94	4.81	6.69	10.01	9.39
	HDD50	5164	983	806	739	414	106	3	0	0	61	358	773	921
	HDD65	9933	1448	1226	1204	864	540	303	208	253	457	821	1223	1386
	CDD50	707	0	0	0	0	30	150	258	213	54	2	0	0
	CDD65	2	0	0	0	0	0	1	1	0	0	0	0	0
	CDH74	2	0	0	0	0	0	2	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.2	5.1	5.7	5.6	4.6	5.4	5.6	5.1	4.4	4.4	4.8	5.6	5.5
Precipitation	PrecAvg	16.4	0.8	0.8	0.7	0.5	0.7	1.1	1.8	2.8	3.0	1.9	1.2	1.1
	PrecMax	27.6	2.7	2.4	2.8	1.9	1.9	3.4	4.5	9.8	7.7	4.3	2.8	2.7
	PrecMin	8.1	0.0	0.1	0.0	0.0	0.0	0.2	0.2	0.3	0.7	0.4	0.0	0.1
	PrecStd	3.5	0.6	0.5	0.5	0.5	0.4	0.7	1.0	1.8	1.7	0.9	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.0	42.1	43.8	52.0	65.8	69.5	70.3	68.3	62.2	56.4	48.0	46.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	38.0	37.9	40.7	48.8	60.1	64.8	66.7	66.0	59.3	51.7	40.5	38.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	34.8	34.5	38.1	46.0	56.8	62.3	64.7	64.0	57.6	49.1	38.6	34.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	32.2	32.3	35.9	43.5	54.3	60.3	63.1	62.1	56.0	46.7	36.3	32.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.8	11.3	11.0	10.7	8.6	7.7	8.4	8.9	8.5	8.4	8.7
		MCDBR	10.0	10.2	10.9	14.0	14.6	11.9	10.3	11.2	9.8	9.7	9.2	10.9
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.234	0.287	0.307	0.348	0.369	0.359	0.372	0.357	0.319	0.298	0.243	0.216
	taud		2.130	2.236	2.271	2.254	2.315	2.423	2.414	2.464	2.558	2.476	2.129	1.943
	Ebn at Noon		160	220	258	267	270	276	268	264	257	220	149	98
	Edh at Noon		12	21	28	35	36	33	32	28	22	16	11	8
All-Sky Solar Radiation	RadAvg		90	309	823	1362	1636	1689	1499	1187	772	359	132	47
	RadStd		13	44	88	119	149	109	132	129	107	43	20	6

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		N/A	+3.60	+3.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page